

APPLICATION FOR APPLYING CORRECTORS WITH COUPLED ERROR TO TOLERANCE

APPLICATION WORKFLOW IN EXCEL

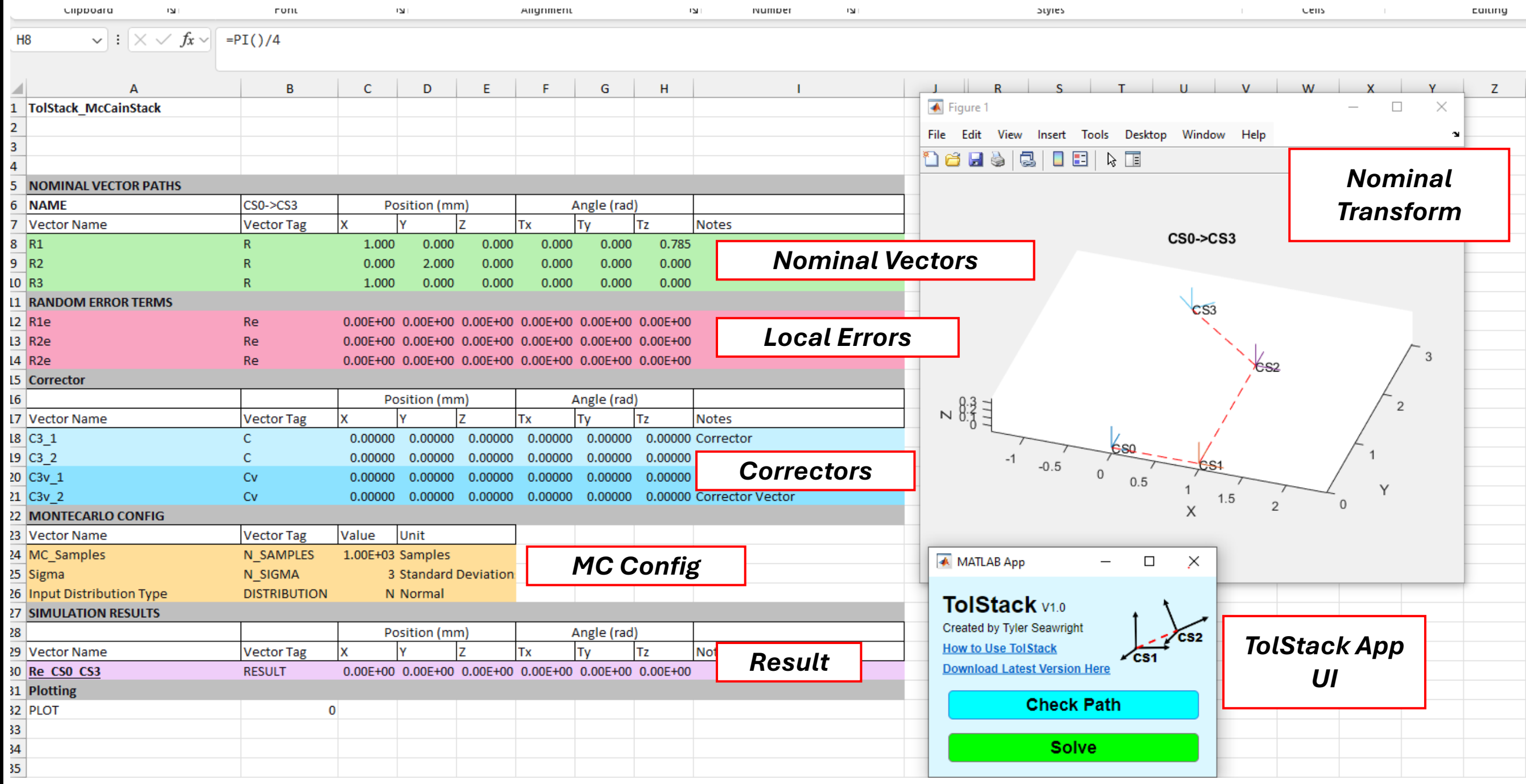


FIGURE 1. The TolStack application is 2 buttons that read from and write to the highlighted cells in an active Excel spreadsheet without any setup or install.

1. Input data with tagged rows for each input.
2. Verify the nominal path with *Check Path*.
3. Output results with *Solve*.

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ERROR CORRECTOR COUPLING AND PROPAGATION

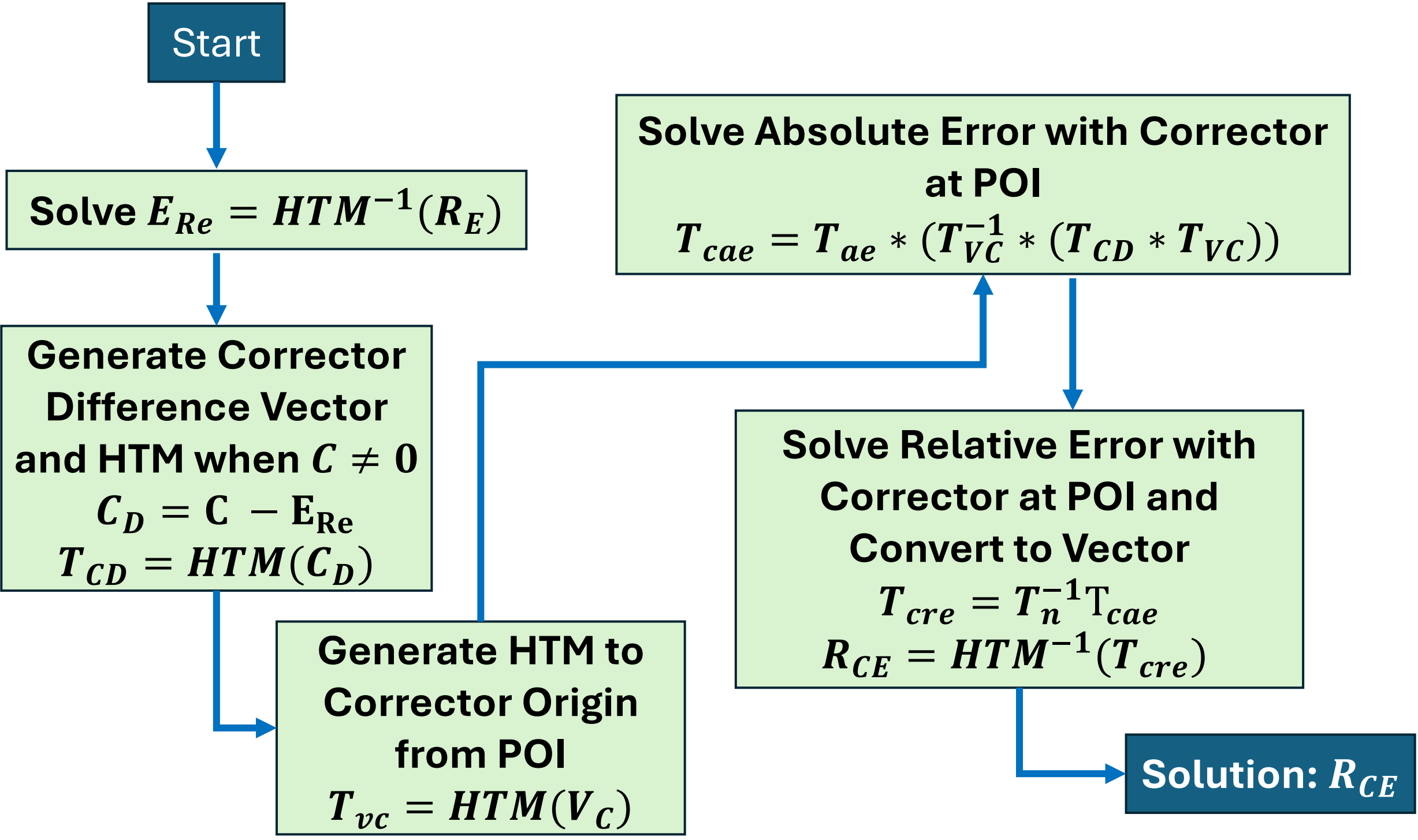


FIGURE 2. Corrector parasitic error is propagated to the system using the corrector difference transform solved by the process in the flowchart.

MODEL VALIDATION 1

Vector Name	X (m)	Y (m)	Z (m)	θ_x	θ_y	θ_z
Rae	0.9999	0.0170	0	0	0	0.0170
Re	-0.0001	0.0170	0	0	0	0.0170
Rce	0	0.0025	0	0	0	0.0025

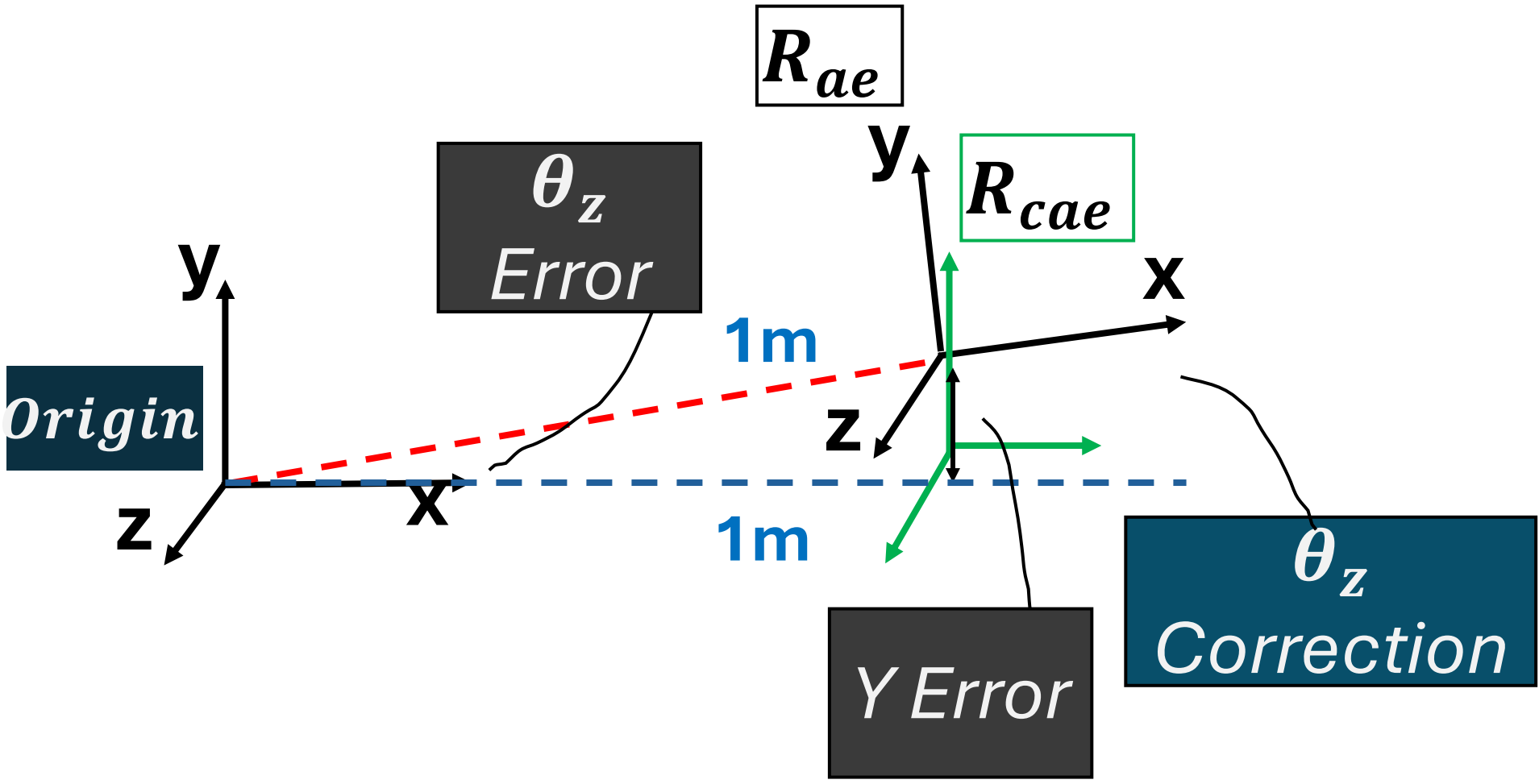


FIGURE 3. A 17mrad θ_z error is corrected to 2.5mrad with the corrector located at the origin.

MODEL VALIDATION 2

Vector Name	X (m)	Y (m)	Z (m)	θ_x	θ_y	θ_z
Rae	0.9999	0.0170	0	0	0	0.0170
Re	-0.0001	0.0170	0	0	0	0.0170
Rce	-0.0004	0.0310	0	0	0	0.0025

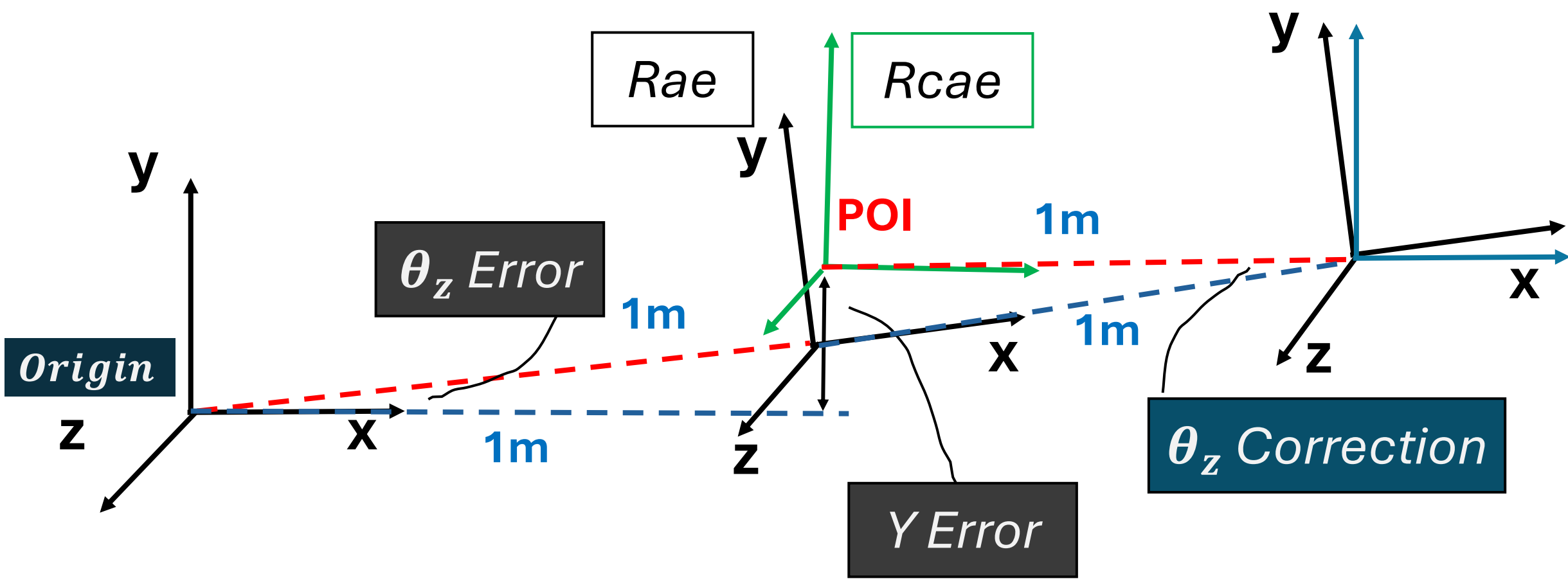
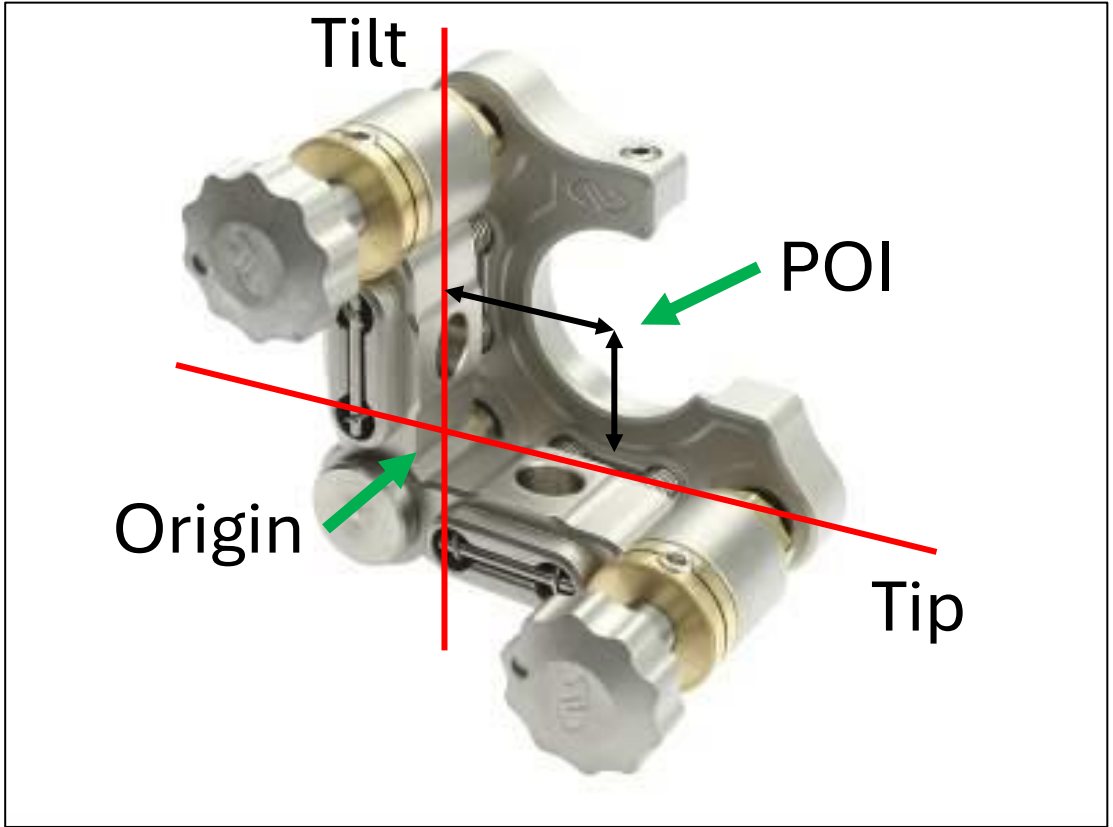


FIGURE 4. A 17mrad θ_z error is corrected to 2.5mrad and adds error in Y direction.

ERROR CORRECTOR DEFINITION



- An error corrector is a device used to correct an error with respect to a reference.

- A corrector at the origin corrects to the resolution of the adjuster with crosstalk.

N FOR APPLYING CORRECTORS WITH COUPLED ERROR TO TOLERANCE ANALYSIS

SAMPLE APPLICATION: OPTIC ALIGNMENT

Task: Determine if the focusing optics of this telescope can be assembled with respect to mechanical datums using mechanical tolerances or require alignment.

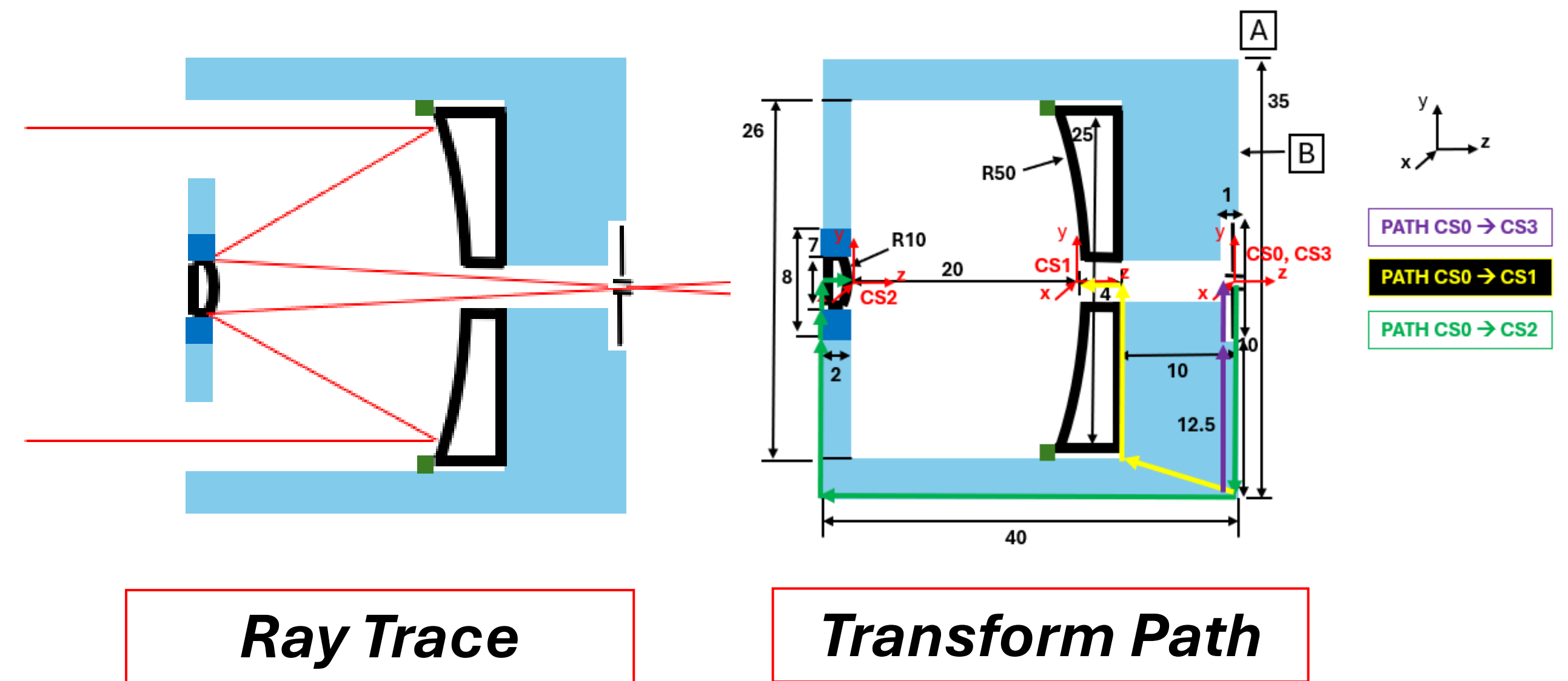


FIGURE 5. Sample optical system under evaluation.

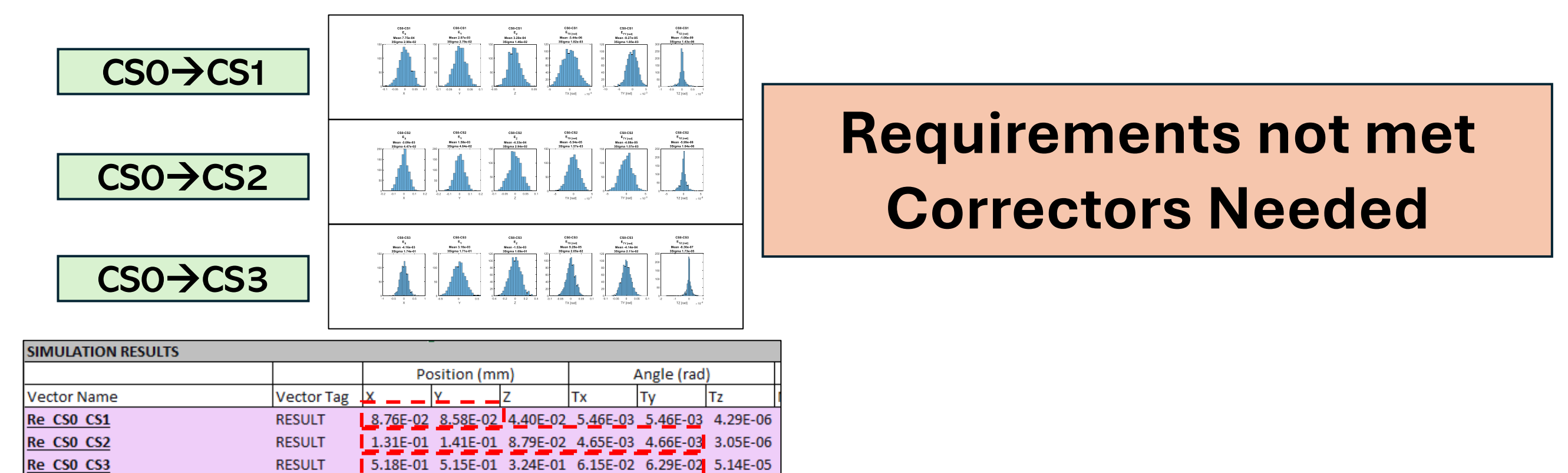


FIGURE 6. Without any correction or adjusters, the requirements are not met so correctors are required to align the optics.

- TolStack output error histograms and $\mu + N\sigma$ error values indicating correctors are needed.

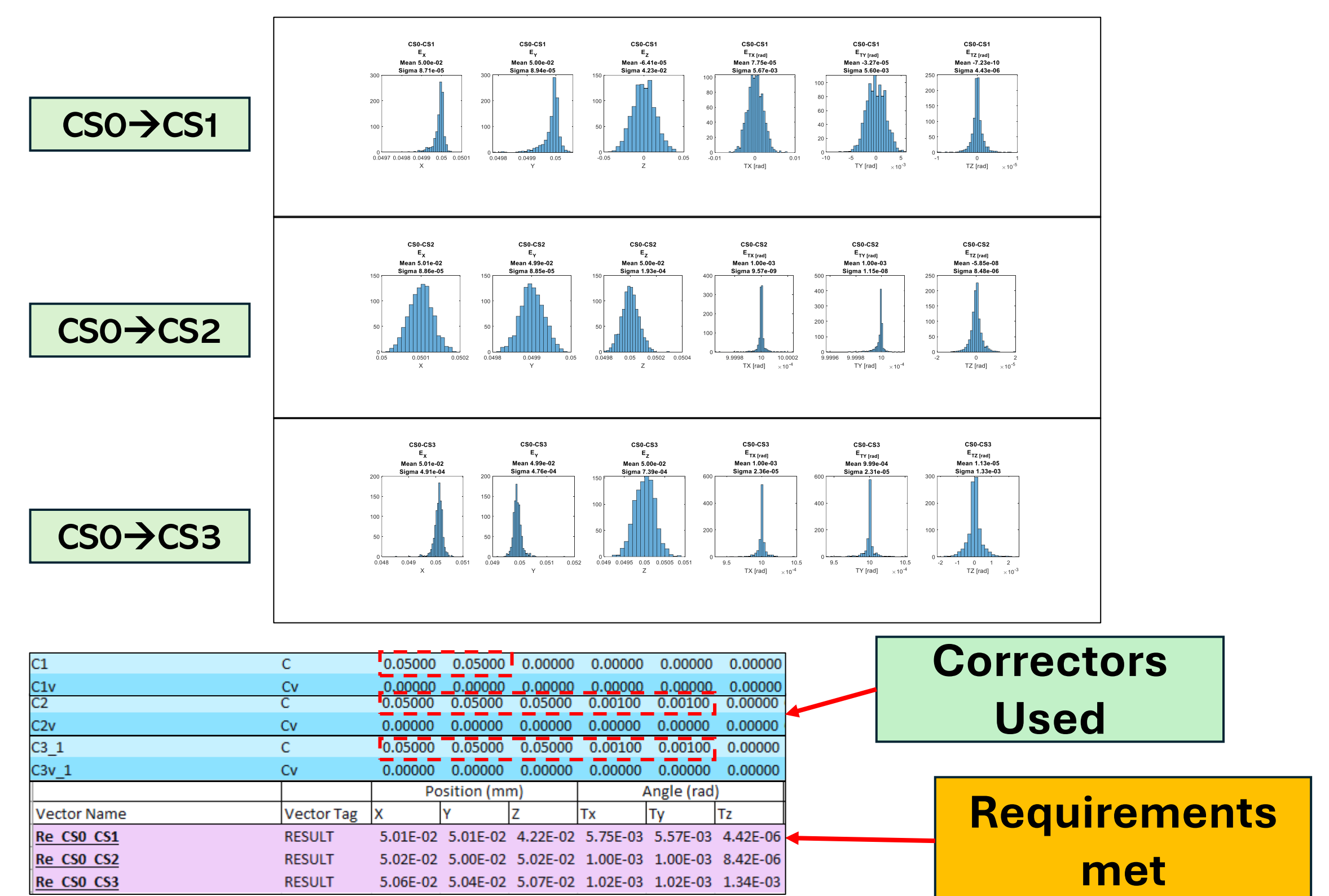


FIGURE 7. Using correctors, the requirements are met and decreased the standard deviation shown by width of the distributions.

- Using correctors, the error is reduced but not eliminated entirely due to coupled motion.

REFERENCES

- 2025 Joint Committee for Guides in Metrology. Evaluation of measurement data – Guide to the expression of uncertainty in measurement. 2008.
- T. A. Seawright, TolStack V1.0 [Computer Software]. 2025. <https://github.com/TylerSeawright/TolStack>
- A. H. Slocum, Precision Machine Design. Prentice-Hall, Inc., 1992.
- Duenner, Andrew C. An Error Budgeting Framework for Engineering Design. Massachusetts Institute of Technology. 2019.